

06 — Graded Year 1 entry sheet (Plan 2.1)

Status: graded run, Year 1 of 4. Fresh \$1 balance sheet: €12,000,000 fixed assets, €54,000,000 cash, €66,000,000 equity, no lines, no stock, no debt. Written 2026-09-06 for a ~60-minute entry session. Every figure traces to a \$ of Scenario MMT39 EN v231.pdf, a test-round result document, or a model/mmt39.py call (the script that produced every model figure below is reproducible from the calls named in the Source column).

Markers: [!] trap · [+] edge · [~] judgement call. No emoji inside tables.

Decisions taken by the team before this sheet was written (stated, not relitigated): all 32 lines installed **and activated** in Year 1, **8 S / 24 H** (changed 2026-09-06 11:50 from 12/20, section 8); fleet sized by tonnage so both legs clear in A and B; loan sized to the real cash trough, not €100M; no factoring; S raw material at production x 1.02 x 1.05, H at production x 1.02 only; **exit E** for Year 1; surveys 20/21/22 + the €11,000 bundle + 3/7/8; selection and training ON; no promotions (S and H both supply-capped, section 8); TV-led media with B matched to A; no SMED in Year 1; no Poka Yoke.

The sheet in ten lines

#	Item	Value	Source
1	Lines	8 S + 24 H = 32 , all installed and activated this year. Capex €52,800,000	\$22; 8 x 2,100,000 + 24 x 1,500,000
2	Shifts / maintenance	3 shifts , preventive maintenance ON, Poka Yoke OFF, SMED 0 (not purchasable in Y1)	\$23, \$25, \$26, \$27
3	Planned output	S 4,646,880 · H 34,851,600 sellable (80.68% of nameplate)	capacity(8, 24, 3, preventive=True)
4	Raw material to buy	S 5,012,280 · H 35,802,000 units, 120-day terms on both	S = 8x3x195,000 x1.02x1.05; H = 24x3x487,500 x1.02 (test-Y1 per line-shift)
5	Prices (MSRP), all channels	S 14.50 in A and B · H 5.25 in A, 7.75 in B · margins S 1.90 / H 0.95 · shelf S 18% / H 20%	Test Y2 DEP p4-p5 (only price points with demand evidence)
6	Sales network	6 managers A, 6 B, 0 E . Reps per zone A: S 4 / T 4; B: S 5 / T 3. Fixed 27,000, variable 0.60%	\$6, \$13; Doc 04 Part 4 #6
7	Fleet	Platforms 12 A / 12 B. Trunk 3 veh A, 2 veh B @ 24,000 kg . Last mile 30 A, 20 B @ 10,000 kg	\$10, \$12; plan_platform_to_retailer(); stockout-flag arithmetic (2.3)
8	Media	Campaign 18 (S) / 19 (H). S: PR 4 + TV 8 per market; H: PR 3 + TV 5 per market. Total €678,000 + POP €48,000	Test Y2 DEP p5-p6, B matched to A
9	Finance	Loan €25,000,000 + credit line €10,000,000 , deposits 0, factoring NO, customer terms 60 days	\$41-\$44; cash trough -19,248,812 in month 2 (section 4)
10	Expected Year-1 net income	€112,983,336 from the model; ~€111,800,000 after three known model corrections (section 4.3)	Simulation.run_year(); test Y2 actual for scale: 87,103,307

[!] Line 10 assumes 100% sell-through of planned output at the Year-2 test prices. The test-Y2 demand readings that support it: S 5,536,395 at 14.50 with B starved of media (3 insertions vs 12 in A); H 35,881,291 at 5.25/7.75. Planned S output 4.65M and planned H output 34.85M are **both below** those readings, so both products should sell out. Unserved demand at those readings: S ~0.89M, H ~1.03M (§34 brand-damage exposure, the smallest of the three mixes compared in section 8).

4-year trajectory (model, base case)

Flat volumes, flat prices, no E, no SMED, loan repaid one third per year. Simulation run with the Year-1 plan below repeated in Years 2-4, surplus cash moved to fixed-term deposits each year keeping a €10,000,000 cash buffer (run basis: section 8).

Year	Lines (S/H)	Units S	Units H	Revenue (MSRP)	Net income	Cash + deposits (31 Dec)	Loan balance	Repaid 2 Jan
1	8 / 24	4,646,880	34,851,600	288,251,775	112,983,336	113,767,230	25,000,000	0
2	8 / 24	4,646,880	34,851,600	288,251,775	115,827,517	229,037,413	16,666,667	8,333,333
3	8 / 24	4,646,880	34,851,600	288,251,775	116,209,271	344,689,351	8,333,333	8,333,333
4	8 / 24	4,646,880	34,851,600	288,251,775	119,933,903	464,065,921	0	8,333,333

Accumulated profit, 4 years: **464,954,028** (the 12/20 mix on the same run basis: 458,008,751). Debt never exceeds 25M against a Year-1 closing cash of 113.8M, so Year-1 borrowing cannot choke Years 3-4. Prepaying the loan in Plan 2.2 out of the Year-1 close (§42, 1% fee) is the obvious Year-2 lever.

Build path and decision points

Year	Capex	Decision gate	Marker
1	32 lines, €52,800,000 (this sheet)	None left: the \$22 cap is reached. Mix 8/24 is locked for four years	[!]
2	2 SMED modules, €6,000,000, if justified	Buy only if the Year-1 REP "Units demanded of your brand" line is > 0 for S or H (unserved demand exists). Model: +14.5% output, +38,977,888 accumulated profit over Years 2-4 if the extra units sell	[~]
2	Poka Yoke: no	12.3-year payback (Doc 00 §3.6); never inside a 4-year game	[+]
3	E: last sensible entry year	Enter only if Year-2 surveys 17/18 show a price point below the local brands and A+B demand is below capacity. A Year-4 entry cannot pay back the 720,000 sales network	[~]
2-4	None otherwise	Everything else on this sheet is a yearly lever (next table)	

Irreversible over four years vs adjustable every plan

Irreversible	\$	Adjustable each year	\$
Lines installed: cannot be removed, 32 total cap, 8/24 mix fixed	\$22	Shifts (1-3), lines activated	\$22, \$23
Poka Yoke (once, all lines)	\$26	Preventive maintenance, training department	\$25, \$32
SMED modules once bought	\$27	Prices, margins, shelf, campaign, media, promotions, POP	\$14-\$21
Brand damage from a finished-goods stockout carries into later years	\$34	Platforms (rented), vehicles, managers, reps, salaries	\$10-\$13
Loan: 3-year commitment, one third leaves cash every 2 January	\$42, \$47	Credit line (1 year), deposits (1 year), surveys, payment terms	\$43, \$44

[!] Sales managers reset to zero every year (Doc 04 Part 6). Re-enter 6/6/0 in every plan.

Form 1 — Commercial distribution

Field	A	B	E	Source
Sales managers	6	6	0	\$6 (50,000 each in A/B). E exit: 0 managers, no E price, no E media (below)

[~] **To enter E instead** (team can overrule): 12 managers \$720,000 + test-Y2 E media \$236,000 + POP 60,000 + 3 trunk vehicles \$158,400 + 12 last-mile vehicles \$432,000 = **\$1,606,400** for the 112,814 units E produced in test Y2 (**\$14.24 per unit**), against three local brands spending \$8,000,000 each on media (survey 12). Source: Doc 05 §6.3 and §7.2; `vehicle_annual_cost()`.

Form 2 — Logistics

Field	A	B	E	Source
Platforms	12	12	12 (form minimum)	\$10 allowed values. 12 is the tested configuration; \$11 platform cost is per order/unit, not per platform
Transport to platforms: number of vehicles	3	2	1 (form minimum)	Section 2.3 below
Transport to platforms: load capacity (kg)	24,000	24,000	1,000 (form minimum)	\$10 max; cheapest per unit at full load, 0.0044 (<code>--vehicles</code>)
Transport to retailers: number of vehicles	30	20	1 (form minimum)	<code>plan_platform_to_retailer(22,246,938, "A", 12, 3815) = 30</code> ; <code>(17,251,542, "B", 12, 2190) = 20</code>
Transport to retailers: load capacity (kg)	10,000	10,000	500 (form minimum)	\$12 max

[!] §13: "Values of zero may not be entered in logistics decisions." E is exited by leaving the E **price** blank (§15: no price = not commercialising), not by zeroing logistics. Test Y1 provisioned 60 platforms and 40 vehicles in E with no E price and no E vehicle cost appeared in the P&L (trunk cost 48,155 total against 86,400 of A+B planned rentals), so the minimum entries above should cost nothing; if charged, 1 x (24,000 + 1,200) + 1 x (24,000 + 600) = \$49,800. Open question 1.

2.3 Why these vehicle counts — the trunk leg is pure arithmetic

One vehicle at load L kg moves 2L units per route, 2,000 / 8 = 250 routes a year (§10). Four test flags confirm it exactly:

Test year / market	Trunk fleet	Capacity units	Units shipped	Flag (REP p4)	Check
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Y1 A	1 @ 17,000	8,500,000	6,340,088	No	[+]
Y1 B	1 @ 15,000	7,500,000	8,089,912	Yes	[+]
Y2 A	2 @ 20,000	20,000,000	20,934,924	Yes	[+]
Y2 B	2 @ 22,000	22,000,000	12,178,692	No	[+]

Planned volume by market (S split 55/45 after matching B media; H split by the test-Y2 sold shares 56.5/43.5): **A 22,246,938 · B 17,251,542.**

Market	Vehicles @ 24,000 kg	Capacity	Headroom over plan	Annual cost	Source
A	3	36,000,000	+62%	158,400	3 x (24,000 + 1.2 x 24,000)
B	2	24,000,000	+39%	105,600	2 x 52,800

[+] Two vehicles would cover A on paper (24.0M vs 22.2M), but test Y2 stocked out at 20.93M on 20.0M of capacity with an identical S/H pattern. The third vehicle is €52,800 against the €27,206,904 of gross profit Doc 05 attributes to the fleet failure.

Last mile has no closed-form rule in §12 (route time "can be any"). Evidence bounds it: test Y2 B stocked out with 10 vehicles @ 10,000 kg on 12,178,692 units, so one vehicle moves **fewer than 1,217,870 units a year**. The model's §8 geometry at 12 platforms gives 30 (A) and 20 (B) vehicles, which loads 741,565 (A) and 862,577 (B) units per vehicle, under the ceiling. Never plan below the evidence floor: A 19, B 15. Cost of 50 vehicles: 50 x 36,000 = **1,800,000**, against the test-Y2 last-mile bill of 2,103,519 that still stocked out in both markets.

Form 3 — Human resources

Field	Value	Source
Level I	126,000	§29 range 108,000-150,000; test Y1/Y2 value
Level II	84,000	§29 range 72,124-99,170; test value
Level III	60,000	§29 range 51,100-72,124; test value
Level IV	35,000	§29 range 30,050-42,075; test value
Operator	13,500	§30 range 12,000-16,830; test value
Specialist	24,000	§30 range 21,000-29,450; test value
Recruitment and training department	Yes	§32, €722,000. Turnover cost 306,000 (Y1) and 276,000 (Y2) with it off

Office payroll at these levels: **2,818,000** (test Y2 REP p5, reproduced by `office_payroll()`). Production headcount at 32 lines x 3 shifts: 648 specialists + 2,424 operators = **3,072**, payroll 48,276,000 (`production_headcount()`, §30).

Sales reps per zone	Channel S	Channel T	Source
A	4	4	Test Y2 (A took 51.9% S / 68.6% H share)
B	5	3	Doc 04 Part 4 #6: B supermarkets are 36-38% of volume, staffed 2/2 in test

Sales force salary	Fixed (€)	Variable (%)	Source
A	27,000	0.60	§13 range 21,000-42,000 / 0.6-1.2; test Y2
B	27,000	0.60	same

Sales network cost: managers 600,000 + reps 2 x 6 zones x 8 reps x 27,000 = **3,192,000** fixed; variable 0.60% of price-to-retailer on normal units in S and T, model estimate 1,191,006.

Form 4 — Market research surveys

Survey	Content	Cost	Select	Source
1	Sales of S by market	3,000	Yes	€11,000 bundle (Doc 00 §3.9)

2	Sales of H by market	3,000	Yes	bundle
3	Sales of S and H by company and market	36,000	Yes	Doc 05 §6.4: only way to compute a rival's share
7	Positioning of S brands	59,900	Yes	Doc 05 §6.4: campaign IDs are meaningless without it
8	Positioning of H brands	59,900	Yes	same
20	Rivals' retail prices	0	Yes	free
21	Rivals' campaigns	0	Yes	free
22	Rivals' promotion types	0	Yes	free
23	Net income by company	1,000	Yes	bundle
27	Platforms by company and market	3,000	Yes	bundle
28	Shifts by company	1,000	Yes	bundle
19	Demand for normal units	110,000	No	REP prints unserved demand free (Doc 05 §5.2)
all others	—	—	No	
Total		166,800		research_cost ([1,2,3,7,8,23,27,28])

Form 5 — Product S: price, margin, shelf, promotions

Market	Channel	M.S.R.P. (€)	Margin (€)	Shelf space (%)	Promotion	Number	Detail	Source
A	G	14.50	1.90	18	No promotion	0	0	Price/margin/shelf: test Y2 DEP p4
A	S	14.50	1.90	18	No promotion	0	0	Promotion: none, S is supply-capped
A	T	14.50	1.90	18	No promotion	0	0	
B	G	14.50	1.90	18	No promotion	0	0	
B	S	14.50	1.90	18	No promotion	0	0	
B	T	14.50	1.90	18	No promotion	0	0	
E	G	blank	blank	blank	No promotion	0	0	\$15: no price = not commercialising

[~] No S promotion (changed with the 8/24 mix): planned S output 4,646,880 is below the test-Y2 demand reading of 5,536,395, so a promotion cannot add sales; §48 rejects promotions the supply cannot cover, and any promo unit that does clear costs 0.40 more than the normal unit it displaces (---promotions 14.50). Revisit if the Year-1 REP shows S demand below output.

POP advertising (€)		A G	A S	A T	B G	B S	B T	E G	Source
S		3,000	3,000	3,000	3,000	3,000	3,000	0	Test Y2 DEP p5; \$19 cap 500,000/channel

Advertising campaign S				A	B	E	Source
Campaign				18	18	0	Test Y2 DEP p5; A took 51.9% S share with it

Advertising repetitions S	A: No.	A: Cost (€)	B: No.	B: Cost (€)	E: No.	Source
DR Display	0	0	0	0	0	
PR Press	4	60,000	4	60,000	0	\$15 rate 15,000; B matched to A (Doc 05 §6.3)
RA Radio	0	0	0	0	0	
SM Social media	0	0	0	0	0	Dearest per reach point, 500 vs TV 300 (---media)
TV Television	8	144,000	8	144,000	0	\$15 rate 18,000; cheapest reach (---media)
Total	12	204,000	12	204,000	0	

Form 6 — Product H: price, margin, shelf, promotions

Market	Channel	M.S.R.P. (€)	Margin (€)	Shelf space (%)	Promotion	Number	Detail	Source
A	G	5.25	0.95	20	No promotion	0	0	Test Y2 DEP p5 (68.6% share in A)
A	S	5.25	0.95	20	No promotion	0	0	
A	T	5.25	0.95	20	No promotion	0	0	
B	G	7.75	0.95	20	No promotion	0	0	Test Y2 DEP p5 (55.1% share in B)
B	S	7.75	0.95	20	No promotion	0	0	
B	T	7.75	0.95	20	No promotion	0	0	
E	G	blank	blank	blank	No promotion	0	0	\$15

[~] No H promotion: planned H output 34,851,600 is already below the test-Y2 demand reading of 35,881,291, and §48 rejects a promotion whose demand cannot be supplied. The H price in A is open question 3.

POP advertising (€)				A G	A S	A T	B G	B S	B T	E G			Source
H				5,000	5,000	5,000	5,000	5,000	5,000	0			Test Y2 DEP p6

Advertising campaign H										A	B	E	Source
Campaign										19	19	0	Test Y2 DEP p6

Advertising repetitions H	A: No.	A: Cost (€)	B: No.	B: Cost (€)	E: No.	Source
DR Display	0	0	0	0	0	
PR Press	3	45,000	3	45,000	0	\$15; test Y2 A and B were equal
RA Radio	0	0	0	0	0	
SM Social media	0	0	0	0	0	
TV Television	5	90,000	5	90,000	0	\$15
Total	8	135,000	8	135,000	0	

Media total S + H: **678,000**. POP total: **48,000**.

Form 7 — Production

Field	H	S	Source
Lines previously installed	0	0	§1 fresh start
Lines installed this year	24	8	§22, 32-line cap; same mix as test Y2, H out-earns S per line (section 8)
Total lines	24	8	
Active lines	24	8	§22: new lines run at full capacity in their first year; A0 silent killer
Stock in units of RM	0	0	§1
Stock in units of FP	0	0	§1
Units purchased of RM	35,802,000	5,012,280	H = 24 x 3 x 487,500 x 1.02; S = 8 x 3 x 195,000 x 1.02 x 1.05

[!] Raw material arithmetic. Test Y1 with preventive maintenance and no SMED produced exactly 195,000 S and 487,500 H per line-shift (REP p4: 780,000 / 4 and 13,650,000 / 28), 1.055x the bare 184,800 / 462,000 plan-on figures; the model's `capacity()` gives 193,620 / 484,050 (81.20%). Scrap consumed material at exactly 1.020000 in Y1 (§34). S carries a further 5% because the test-Y2 S shortfall of 442,600 units was sourced at a premium (Doc 05 §7.1); H carries none because the test-Y1 over-order locked €109,431,358 (Doc 04 §1.5). Raw material cost at list: S 8,270,262 + H 37,592,100 = **45,862,362** (§49 rates 1.65 / 1.05).

Production options	Value	Source
Number of shifts	3	§23; depreciation 12% (§35)
Preventive maintenance	Yes	§25, 500,000; +2.5% output on 32 lines (80.68% vs 78.75%)
Pokayoke previous years	No	
Pokayoke	No	§26; 12.3-year payback (Doc 00 §3.6)

SMED	Value	Source
SMEDs previous years	0	
To acquire	0	§27: not available before year 2
Total SMED	0	
Active	0	
Teams assigned to SMED per shift	0	

Standard cost the model expects at these settings: **S 5.3217 · H 2.1718** (`standard_cost()` with preventive, salaries 24,000 / 13,500). Test Y1 actuals on the same configuration at two shifts: S 5.5371 / H 2.1788.

Form 8 — Finance

Field	Value	Source
Loan requested	25,000,000	Section 4: month-2 trough -19,248,812 + 5,751,188 buffer (29.9%)
Loan prepayment	0	
Financial investment	0	§44: 1-year lock, cashed 2 Jan next year; every euro is needed by February
Fixed-term deposits	0	same; the deposit lever starts in Plan 2.2 with the Year-1 closing cash
Line of credit requested	10,000,000	§43: 1% fee = 100,000 as insurance against the 20% overdraft (§42) if sales slip
Customer payment terms	60 Days	§38 shortest option; receivable 48,041,962 at year end (2/12 of revenue)
Factoring	No	Team decision; test Y2 paid 1,995,715 to factor receivables that closed at zero
Max. amount to be discounted	--	
Payment term for suppliers S	120 Days	§28; test Y2 S raw material at 120 days came in at list 1.65 (Doc 05 §7.1 arithmetic)
Payment term for suppliers H	120 Days	§28; payable at year end 4/12 x 45,862,362 = 15,287,454 of free financing
Investment in securities 1-3	0 / 0 / 0	§45
Sale of shares 1-3	No / No / No	

4. Cash check

4.1 The §41 test the simulator runs

Component	€	Source
80% x new fixed assets 52,800,000	42,240,000	§41 item 1
50% x (media 678,000 + POP 48,000 + managers 600,000 + rep fixed 2,592,000 + overhead 600,000 + research 166,800)	2,342,400	§41 item 2
15% x raw material 45,862,362	6,879,354	§41 item 3
OCN	51,461,754	<code>operating_cash_needed()</code>
Available cash 2 January	54,000,000	<code>available_cash_jan2(54,000,000)</code> : no payables, no loans due

Headroom	+2,538,246	<code>financing_gap()</code> -> <code>loan_needed</code> 0
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[!] The OCN test passes, so the simulator will **not** borrow for us. That is not the same as having enough cash: \$37 says lines are paid in cash, so on 2 January cash is 54,000,000 - 52,800,000 = **+1,200,000**, less than one month of payroll, and the first 60-day receipts arrive in March.

4.2 The real trough (monthly, before financing)

Outflows spread evenly (model income lines, 8/24 run): conversion cost 55,870,240 (production payroll 48,276,000 plus power and repairs), other operating cost 24,884,539 (office 2,818,000, overhead 600,000, preventive 500,000, training 722,000, sales network 3,192,000, media 678,000, POP 48,000, research 166,800, logistics 3,188,409, R&D 12,971,330), retailer margins 41,938,092 paid at sale (\$38). Raw material 45,862,362 paid from month 5 (120 days). Revenue 288,251,775 received from month 3 (60 days). Capex 52,800,000 on 2 January.

Month-end	Jan	Feb	Mar	Apr	May	Dec
Cash	-9,024,406	-19,248,812	-5,452,236	8,344,339	18,319,051	88,142,034

Financing	€	Cost in Year 1
Loan	25,000,000	2,500,000 interest + 250,000 fee (\$49: 10% + 1%)
Credit line	10,000,000	100,000 fee; 7% only on any draw (\$43)
Facility vs trough	35,000,000 vs 19,248,812	1.8x cover; a 30% shortfall in March receipts still clears
Deposit	0	nothing to park until the Year-1 close

Test comparison: test Y1 with no planned loan cost 1,892,199 of overdraft on ~9.46M average; test Y2 with a 100,857,687 loan cost 13,977,015 of financial expense while closing with 107,327,322 of cash.

4.3 Year-1 result from the model and the year-end position

`simulation.run_year()` with this sheet: revenue 288,251,775; net sales 246,313,683; gross margin 145,893,875; operating income 113,233,336; **net income 112,983,336**. Year-end: cash 113,767,230, receivables 48,041,962, payables 14,849,856, loan 25,000,000, inventory 0. Available cash for the Year-2 OCN test: **138,626,003**, after the 8,333,333 repayment.

Two model behaviours the test round has already measured, applied by hand rather than by editing the model:

Correction	Evidence	Effect on Y1 net income
Loan interest is charged in the draw year	Test Y1: "Loans 382,605" = 10% x 3,826,050 drawn that year; model charges 0	-2,500,000
Credit-line fee	Test Y2: 300,000 = 1% x 30,000,000, undrawn	-100,000
Land and buildings are not depreciated	Test Y2: 7,056,000 = 12% x (52,800,000 lines + 6,000,000 SMED); model applies 12% to 64,800,000	+1,440,000
Adjusted Year-1 net income		~111,823,000

[~] For scale: test Y2 earned 87,103,307 on 254,505,471 of revenue with 12,253,570 H units unsold and 13,977,015 of financial expense. This plan has 13% more revenue, one fifth of the financial expense, and no unsold stock **if the fleet clears** - which is the whole bet.

5. A6 pre-submission checklist, ticked against this sheet

Check	Status	Where
Price, retailer margin and shelf space in every cell we sell in	Yes	Forms 5-6: 12 A/B cells; E deliberately blank
Advertising campaign selected	Yes	18 S / 19 H in A and B
Lines activated, not just installed	Yes	Form 7: 20 H + 12 S active
Production forecast <= capacity at the plan-on rate	Yes	<code>capacity_reality_check</code> : S 4,646,880 / H 34,851,600, no STOCKOUT flag
Raw material covers production plus scrap	Yes	H x1.02; S x1.02 x1.05

No logistics field blank or zero	Yes	Form 2, E at form minimums (open question 1)
Every zone covered by a manager or wholesaler	Yes	6/6 managers in A/B; E not operated
Promotion volumes within supply	n/a	No promotions: both products are supply-capped (\$48)
OCN <= available cash, or loan deliberately requested	Yes	51,461,754 <= 54,000,000 and 25M loan for the trough
Surplus cash on deposit	n/a	No surplus until 31 Dec; deposit in Plan 2.2
Surveys 20, 21, 22 + the 11,000 bundle ordered	Yes	Form 4, total 166,800
SMED not requested in year 1; Poka Yoke not requested	Yes	Form 7
Sales managers entered (they reset every year)	Yes	Form 1

6. Open questions to confirm on the interface (max 5)

#	Question	If the answer is bad
1	Does the E logistics block accept the form minimums (12 / 1 @ 1,000 / 1 @ 500) with no E price , and does the DEP mirror show E as not commercialised? §13 forbids zero; §15 says no price = not selling	If the form forces a price, enter one above the E average (test Y2: 19.50 S / 9.00 H) and 0 shelf space so no E sales occur
2	Graded-run INF: loan maturity and rate. Test went 4 -> 3 years; §49 says it keeps falling	At 2 years the repayment is 12,500,000 on 2 Jan Y2 and Y3, against 108.6M closing cash: still fine, update <code>SCENARIO</code>
3	H price in market A: hold 5.25 (68.6% share evidence) or lift toward B's 7.75 (55.1% share evidence)? Planned H output 34.85M is 1.0M below the test-Y2 demand reading, so a little H demand goes unserved either way (§34)	Doc 05 §7.3 says re-read price only once the fleet is proven; if the team lifts A, do it in steps that keep the B-market evidence as the ceiling
4	Promotion form semantics: is "Detail" the gift value and "Number" the count of promotions (test Y2 E entry 2 / 2 cost 0.60 per promo unit, 51,428 total)?	If "Detail" is a euro budget, set it to 0.40 x expected promo units per channel (--- <code>promotions 14.50</code>)
5	Last-mile count: the interface may show a per-vehicle capacity or route time. If it does, resize: units per vehicle must stay above 20,374,889 / A-count and 15,895,111 / B-count	Never below A 17 / B 14 (the test-Y2 evidence floor of 1,217,870 units per vehicle)

7. Where Doc 05's plan did not translate to a fresh start

Doc 05 item	Fresh-start translation
#3 "Sell the H you already made, buy no H raw material, idle H lines"	No stock exists. H raw material must be bought (35,802,000) and every line activated; the 0.9379 cheap-material block is gone, so H standard cost is 2.17, not 2.06
#2 "Right-size the debt vs 100M, deploy the cash"	There is no 100M loan to right-size and no payables or loans due on 2 January; the loan is a fresh 2-month trough loan, and there is no surplus to deposit until the close
Part 4 calibrations (SMED 92.9%, conversion cost -6%/-8%)	Both are SMED-plant measurements; Year 1 has no SMED, so the model correctly runs the bare-line 81.20% and uncalibrated costs
#1 "Fix the fleet" from stockout flags	Translated into tonnage arithmetic (2.3) because there are no Year-0 flags to react to
#4 "Rebalance toward S"	Reversed for the fresh start. Doc 05's case for S rested on H being unshippable. With the fleet sized to clear, H earns more per line-shift (Doc 05 #4: 1.797M vs 1.634M) at 1.5M capex vs 2.1M. Section 8
§7.2 "Decide E explicitly"	Decided: out. Entry cost stated on Form 1 for the overrule
4-year horizon (new requirement)	All capex lands in Year 1 by team decision; Years 2-4 have one gated capex (SMED) and no line additions possible

Figures that could **not** be sourced and are therefore judgement calls: the 55/45 S market split (interpolated between test Y1 48/52 and test Y2 98/2), the 15% loan buffer, the €10,000,000 credit-line size, and the last-mile vehicle counts beyond the evidence floor.

8. Line mix: why 8/24 and not 12/20 (decided 2026-09-06 11:50)

The 12/20 mix was carried over from Doc 05 item 4, whose argument was that the marginal H line-shift was worth zero because the test-Y2 fleet could not ship it. This sheet sizes the fleet to clear both legs, so that argument no

longer applies, and the choice reverts to earnings per line: H out-earns S per line-shift and costs 1.5M against 2.1M to install. `Simulation.run_year()` with this sheet's plan (same prices, media, fleet, terms, loan; sales lines A/B 55/45 for S and 56.5/43.5 for H; last-mile cost scaled to volume), with and without capping sales at the test-Y2 demand readings (S 5,536,395, H 35,881,291):

Mix (S/H)	Capex	Net income, all output sold	Net income, S capped at 5,536,395	Unsold S	H demand unserved
12 / 20	55,200,000	111,349,089	102,368,855	1,433,925	6,838,291
10 / 22	54,000,000	112,571,121	110,789,519	272,205	3,933,991
8 / 24	52,800,000	112,983,336	112,983,336	0	1,029,691

[+] 8/24 wins in both cases, needs 2.4M less cash on 2 January, leaves the least demand unserved (\$34), and the mix is locked for four years (\$22). The 12/20 headline in the earlier version of this sheet (110,187,386) was the same plan on a slightly different run basis; the like-for-like figures are the ones in this table. What changed on the forms: Form 7 lines and raw material, Form 5 S promotion (removed, S is now supply-capped), Form 2 volumes (vehicle counts unchanged per the model), and every cash figure in section 4. Nothing else moved.

[~] Judgement in this: the demand readings are one test year at one media split. If the Year-1 REP shows S demand well above 4.65M and H below 34.85M, the SMED gate in Year 2 is the only lever left.

Sources

- Scenario MMT39 EN v231.pdf — \$1, \$6, \$8, \$10-\$15, \$18-\$23, \$25-\$30, \$32, \$34, \$35, \$37, \$38, \$41-\$45, \$47-\$49
- day 1/year 2/DIR0114SI02082TE3Y1REP.pdf — units by channel p2-p3, production and stockout flags p4, P&L p5, balance sheet p6
- day 1/year 2/DIR0114SI02082TE3Y1DEP.pdf — Year-1 form layout and values, logistics p2, HR p3, production p6, finance p7
- day 2/year 2 results/DIR0114SI02082TE3Y2DEP.md — form order and field names used above
- day 2/year 2 results/DIR0114SI02082TE3Y2REP.md — units by channel p2-p3, production and flags p4, P&L p4-p5, balance sheet p6
- day 2/year 2 results/DIR0114SI02082TE3Y2INFen_US.md — loan maturity 3 years, rates
- docs/03-decision-checklist-and-grading-playbook.md — A0-A6
- docs/04-test-year1-postmortem-and-year2-audit.md — Part 1 failures, Part 3 form template, Part 4 gap list
- docs/05-test-year2-postmortem-and-graded-run-plan.md — Parts 1, 2, 4, 6, 7
- model/mmt39.py — `capacity()`, `standard_cost()`, `production_headcount()`, `office_payroll()`, `plan_platform_to_retailer()`, `vehicle_annual_cost()`, `platform_costs()`, `operating_cash_needed()`, `available_cash_jan2()`, `financing_gap()`, `research_cost()`, `promotion_menu()`, `Simulation.run_year()`, `LoanBook`